

Muhammad Hassan

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Engineering student with hands-on experience in computer vision, deep learning, image processing, and Python-based ML pipelines. Built projects spanning a 9-stage automated attendance and monitoring pipeline, white blood cell enhancement and classification, and real-time YOLO weapon detection. Strong foundation in data structures, algorithms, and neural network workflows, with AWS Cloud Foundations certification.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, TypeScript
AI / ML	Machine Learning, Deep Learning, Computer Vision, YOLO, TensorFlow, OpenCV, Image Processing
Pipelines	Data Preprocessing, Annotation, Augmentation, Model Training, Evaluation, Deployment Workflow
Web / Mobile	React, React Native
Tools & Platforms	GitHub, Jira, Notion, n8n, AWS
Certifications	AWS Academy Graduate - Cloud Foundations

WORK EXPERIENCE

Internee | *Nexentra Solutions* | Jun 2025 - Oct 2025

- Contributed to mobile application development using React Native while strengthening practical software engineering workflows.
- Collaborated with senior engineers on feature implementation and consistently delivered assigned tasks ahead of schedule.
- Demonstrated adaptability by independently learning new tools and frameworks within project timelines.

EDUCATION

B.Sc. Computer Engineering | *Information Technology University, Lahore* | Aug 2022 - Jun 19, 2026

- CGPA: 3.25 / 4.0

Relevant Coursework:

- **Programming Fundamentals, Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Machine Learning, Computer Vision**

PROJECTS

Automated Attendance & Monitoring System - *Python, OpenCV, Computer Vision, Machine Learning* | Final Year Project

- Implemented a 9-stage Python pipeline for automated attendance capture, monitoring, and reporting workflows.
- Applied computer vision techniques for preprocessing, detection, recognition, validation, and attendance logging.
- Designed the pipeline to reduce manual attendance effort and support reliable monitoring in academic environments.

White Blood Cell Enhancement & Classification - *Python, OpenCV, Deep Learning, Computer Vision* | Dec 2024

- Designed and implemented an end-to-end image processing pipeline to enhance and classify white blood cell types from microscopy images.
- Applied a deep learning classification model to support accurate identification across multiple cell categories.
- Conducted at Information Technology University as a research-oriented academic project.

Real-Time Weapon Detection System - *Python, YOLO, Machine Learning* | Jan 2025

- Built a real-time weapon detection system using YOLO object detection for live video stream processing.
- Trained and fine-tuned the model on custom datasets, optimizing detection speed and accuracy for security applications.
- Demonstrated a full ML workflow covering data collection, annotation, training, evaluation, and deployment preparation.

ADDITIONAL INFORMATION

Languages: English (Fluent), Urdu (Native), Punjabi (Native)

Interests: Computer Vision Research, AI for Healthcare, Autonomous Systems